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Limitations

Our study on the Depth Up-Scaling (DUS) has important limitations and considerations. One key limitation is the need for more thorough explorations of hyperparameters used in the DUS approach. Namely, we removed $m = 8$ layers from both ends of our base model, primarily due to hardware limitations. However, we have not yet determined if this value is optimal for enhancing performance. The extended time and cost of continued pretraining made it challenging to conduct more comprehensive experiments, which we aim to address in future work through various comparative analyses.

In terms of the model's broader implications, there are several points to note. The model's significant computational demands for training and inference might limit its use, especially for those with restricted computational resources. Additionally, like all machine learning models, it is vulnerable to biases in its training data, which could lead to skewed outcomes in certain situations. Furthermore, the substantial energy consumption required for training and operating the model raises environmental concerns, which are critical in the pursuit of sustainable AI development.

Lastly, while the fine-tuned variant of the model shows improved performance in following instructions, it still requires task-specific fine-tuning for optimal performance in specialized applications. This fine-tuning process can be resource-intensive and not always effective. Recognizing and addressing these limitations is essential for a comprehensive understanding of the proposed Large Language Model's capabilities and for guiding future research and development in the field of LLMs.

Ethics Statement

We conscientiously address and emphasize the commitment of SOLAR 10.7B in maintaining the highest ethical standards. First, we highlight that SOLAR 10.7B-Instruct has shown low levels of data contamination in our evaluations, a testament to our rigorous data handling and processing protocols. This aspect is crucial, as it underpins the reliability and integrity of the results obtained from SOLAR.

Furthermore, during the course of our experiments, we ensured that all setups and methodologies employed steer clear of any potential ethical pitfalls. This preemptive consideration and avoidance of ethically questionable practices underscore our dedication to conducting research that is not only innovative but also responsible.

Additionally, we ensure that SOLAR complies with general ethical considerations in all aspects of its operation. This includes adherence to privacy norms, respect for intellectual property, and ensuring the absence of bias in our algorithms. Our commitment to these ethical principles is unwavering, and we believe it significantly contributes to the credibility and societal acceptance of SOLAR.

In conclusion, the ethical framework within which SOLAR operates is robust and comprehensive, ensuring that our advancements in this field are not only scientifically sound but also ethically responsible.

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